# Lecture II - First Steps in Julia

Applied Optimization with Julia

Dr. Tobias Vlćek
University of Hamburg

# Technical Setup

# Goals for Today

After this lecture, you will have:

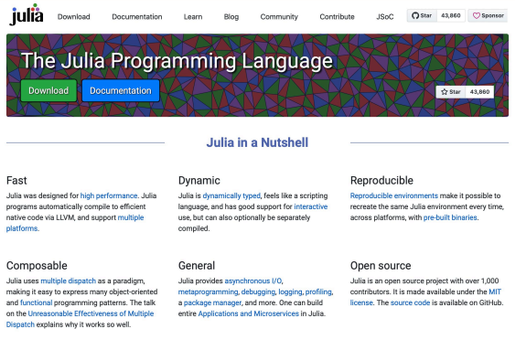
- A **working Julia and Zed setup** on your computer
- Made your first Git commit
- Learned how to submit assignments and earn bonus points
- Started with the first tutorials

# Download and Install Julia




To prepare for the upcoming lectures, we start by installing the Julia Programming Language and an Integrated Development Environment (IDE) to work with Julia.

# Installing Julia




- Head to julialang.org/downloads and follow the instructions
- macOS/Linux: run the install command in a terminal
- Windows: download and open the **MSIX App Installer**
- Later, you can then manage Julia with `juliaup`



If you are ever asked to add something to your “PATH”, do so!

# Zed

- Next, we are going to install Zed, the editor of this course
- macOS: `brew install --cask zed` (or the installer from zed.dev)
- Windows: installer from zed.dev/download
- Linux: `curl -f https://zed.dev/install.sh | sh`



On Windows, Zed needs a DirectX 11 GPU with a current graphics driver. If it shows a blank window or an “Unsupported GPU” warning, update the driver first. Still not working? Use VS Code with the “Julia (Julia Language Support)” extension instead: everything in this course works there too.

# Verify the Installation

- Start Zed and take a look around
- Open the command palette: `Ctrl+Shift+P` (Windows/Linux) or `Cmd+Shift+P` (macOS)
- Choose **zed: extensions** and search for “Julia”
- Install the Julia extension for syntax support

# Create a New File

- Create a new file with a ".jl" ending
- Save it somewhere on your computer
- e.g., in a folder that you will use in the course

```
1 println("Hello World!")
```

```
Hello World!
```

# Run the File

- Open a terminal in the folder where you saved the file
- Run it with `julia test.jl`
- Later, we will run files **cell by cell** directly from Zed — the first tutorial sets that up

# Everything Working?

- If the terminal prints `Hello World!` → perfect!
- If not, the terminal likely **cannot find the path** to Julia
- Open a new terminal after installing Julia (restart Zed if you use its terminal)
- Then try to run the file again

> Don’t worry if it is not running right away. We will fix this together! On Windows, the FAQ lists the usual fixes.

# Learning Julia

# Julia as a Language

- The following three lectures are dedicated to learning the basics
- Start with the very basics and gradually move on
- Focus in the first two lectures on the programming language
- Third lecture dedicated to **Mathematical Optimization**

# A First Taste of Julia

## Question: What will your savings be worth in 10 years?

```julia
savings = 1000        # Euros in your account
interest_rate = 0.03 # 3% interest per year
value = savings * (1 + interest_rate)^10
println("Value after 10 years: ", round(value; digits=2))
```

```
Value after 10 years: 1343.92
```

- Don’t worry if this looks new — you will be able to **read and write** code like this after the tutorials!

# Your Turn

Build the turbines with the most energy: tap a site. Bigger turbines give more, but no turbine may stand in another's wind shadow. Add sites to see how long a loop that tries every combination would need, at a billion per second.



+10 sites

$2^{10}$ combinations: a loop needs **1 µs**

You **0 GWh**

Ready: press Optimize to compare with the best plan.

Optimize

Reset

New

# Julia in Zed

# Tutorials in Zed

- The easiest way to work on the tutorials is a `.jl` file in Zed, run cell by cell
- For detailed instructions, just open the first tutorial
- It explains step-by-step how to set up the course environment and run the cells

> ⓘ If you use `.jl` files, you can also put them under version control with Git, as you will see later in this lecture.

# Downloading the Tutorials

- You will find the tutorial files next to the tutorial pages
- On each page, you will find a button `Julia` on the right
- Click it to download the `.jl` file and save it
- The file is split into cells by `# %%` markers, ready to run cell by cell
- Store them **in the `applied-optimization` folder** that the first tutorial creates

# Learning by Doing

- The best way to learn a programming language is **by doing**
- We will therefore solve problems the coming weeks
- The goal is to get you familiar with the language
- You can discuss the problems with your fellow students
- You can hand in your solutions to receive bonus points!

# Working with Git

# What is Git?

- Git is a **version control system** that tracks changes in your code
- Helps you collaborate and keep track of your work
- Saves “snapshots” of your project at different stages
- You can always go back to previous versions if something goes wrong
- No need to create files like `tutorial_v1.jl` and `tutorial_v2.jl`

# Installing Git

- Head to git-scm.com and download Git
- Follow the installation instructions on the website for your OS



If you have any questions, feel free to ask!

# Git in Zed

- Zed has **built-in Git support**!
- Open the **Git panel** with the Git icon in the status bar
- Staging, committing and file history need no extension
- Windows: install Git for Windows first, then restart Zed



Git is not required for the course, but strongly recommended: once you get used to it, it becomes invaluable, especially if you are working with a lot of code!

# Initialize a Repository

- Open your project folder (of our lecture) in Zed
- Open the Git panel and initialize a repository
- Or run `git init` in a terminal in that folder
- Your folder is now a Git repository!



You can also synchronize your repository with GitHub or other hosting services. Then, your code is saved in a remote location, making it accessible from anywhere and allowing collaboration with others.

# Making Your First Commit

- Make changes to your files (e.g., work on a tutorial `.jl` file)
- Open the Git panel
- You'll see your changed files listed there
- Tick the checkbox next to files to **stage** them
- Add a commit message describing your changes
- Press `Cmd+Enter` (macOS) or `Ctrl+Enter` to **commit**

# Viewing History

- Right-click a file in the file list and choose View History
- See every commit that changed this file
- Select a commit to see what changed in it



Start using Git from day one! Even for small projects, it’s a good habit to develop.

# Assignments

# Find Your Group

- You hand in the assignments as a **group of up to three**
- Take a few minutes now: who do you want to work with?
- Tell me your group today and I will set it up in Moodle
- Your group stays the same for the whole semester



You would rather work alone? That is fine too: you are then a group of one.

# Submission of Assignments

- Submit the assignment via Moodle: one upload counts for your group
- Upload the tutorial `.jl` files and keep their file names
- The assignment is due **at the start of the next tutorial**
- After that, Moodle does not accept uploads anymore



Don’t forget to save your files before uploading them to Moodle!

# Grading of Assignments

- Each assignment gives up to 5 points in Moodle
- Divided by 10, that is up to 0.5 bonus points for the exam
- You can get a maximum of 6.0 bonus points from the assignments
- You need to pass the exam first, to receive any bonus points!



The assignments are **not** mandatory, but highly recommended!

# What About AI?

- You may use AI tools: say so in one line in your submission
- But the bonus points only count if you pass the exam
- And in the exam, there is no AI to help you
- So it pays off to **solve the tutorials yourself**

> ⓘ The full rules are in the AI policy on the course website.

# Five Tutorials

# Topics of the Tutorials

- Variables: Learn how to assign values to variables
- Vectors: Learn how to create and manipulate vectors
- Comparisons: Learn how to compare values
- Loops: Learn how to use loops to repeat code
- Dictionaries: Learn how to store and look up key-value pairs



Dictionaries will become important later: they are how we will store and look up the data of our optimization models.

# Get Started with the Tutorials

- Download the first tutorial file and open it
- Start with the **first problem and solve it step by step**
- You can find the tutorials here on the website
- **You can ask questions anytime!**

> ⓘ And that's it for this lecture!
>
> The remaining time we will already start working on the first problems.

# Literature

# Literature

- Lauwens, B., & Downey, A. B. (2019). Think Julia: How to think like a computer scientist (First edition). O'Reilly®. Link to the free book website.
- Julia Documentation

For more interesting literature to learn more about Julia, take a look at the literature list of this course.